

Qatar-1b

R-2785

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140802 · created 2026-09-04T00:27:39+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456871.69377 +/- 0.00035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.057
Transit depth (Rp/Rs)^2	3.6 +/- 2.15 [%]
Orbital Inclination (inc)	83.26 +/- 2.8
Airmass coefficient 1 (a1)	0.155 +/- 0.071
Airmass coefficient 2 (a2)	1.47 +/- 0.72
Scatter in the residuals of the lightcurve fit is	41.14 %
Best Comparison Star	#1 - [24, 105]
Optimal Aperture	2.76
Optimal Annulus	4.75
Transit Duration (day)	0.061 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.19 $\pm$ 0.057 vs 0.1463 (deviation 0.77 σ)
Duration fit vs published	0.061 d vs 0.0692 d (deviation 0.35 σ)
Mid-time O-C	-0.5 min
Depth SNR	3.3
Residual scatter	41.14 %

Light curve

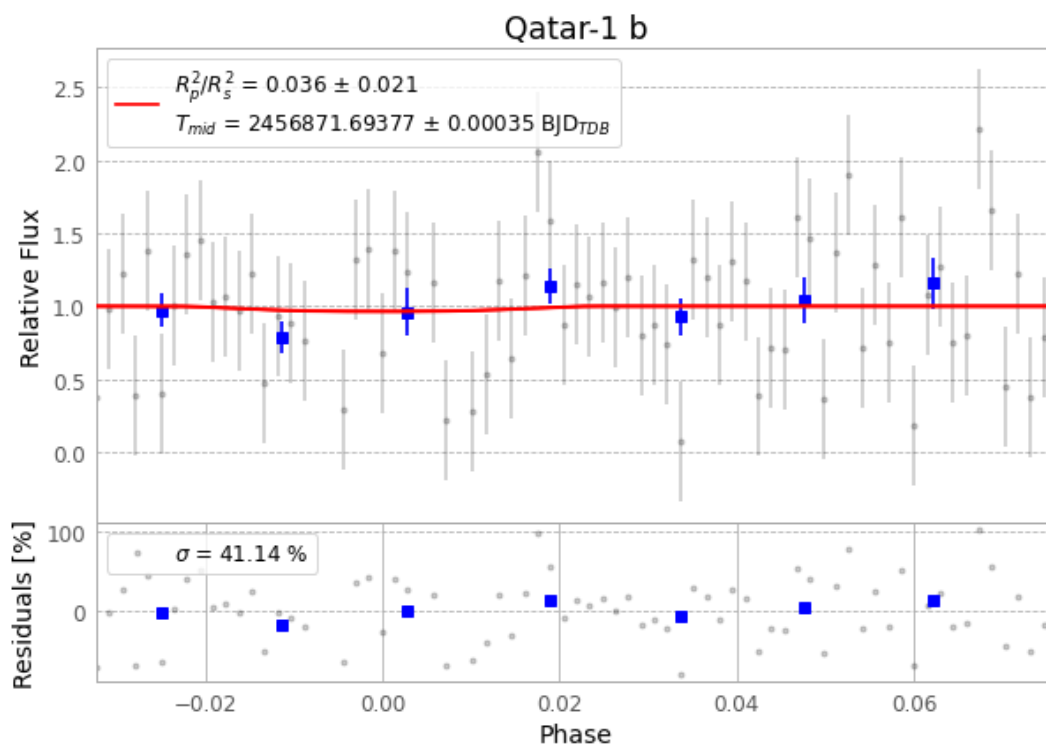


Figure 1 — FinalLightCurve_Qatar-1 b_2014-08-01.png (SHA-256 122e58b7b7c9d476...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [561, 313], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9373a7e16bb36ea2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

75 FITS frames (49 MB), Qatar-1140802033012.FITS → Qatar-1140802071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140802.log (b9ec4ee66672...), FinalLightCurve_Qatar-1 b_2014-08-01.png (122e58b7b7c9...), FinalLightCurve_Qatar-1 b_2014-08-01.csv (00ed065d9176...)

Compute resources

Wall time 131.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 00:27Z.